
DRONE DELIVERY DATA ANALYSIS

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DRONES CONTEXT



→ Used for making deliveries by companies such as:

- ◆ Amazon
- ◆ Walmart

→ The speed of these drones may be affected by factors such as:

- ◆ Weather
- ◆ Package weight
- ◆ Traffic
- ◆ Battery Level
- ◆ Route complexity

MODEL PURPOSE



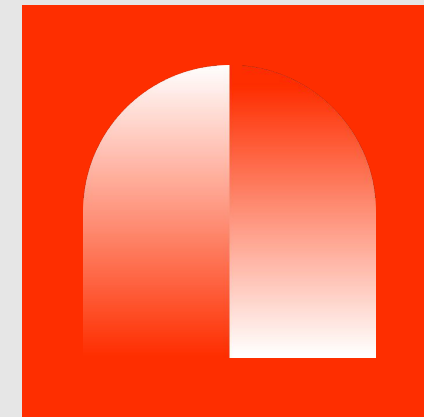
- This model analyzes the effect of these factors on the speed of the drone
- This analysis determines the conditions that are most and least impactful

DATA MINING METHOD



Data Collection

We used the given data on Moodle.



Data Preparation

- *Filtered out any robot deliveries out data*
- *Made sure all the deliveries were successful*

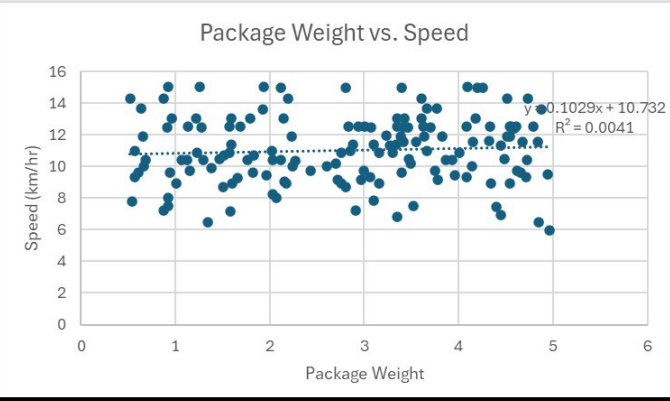
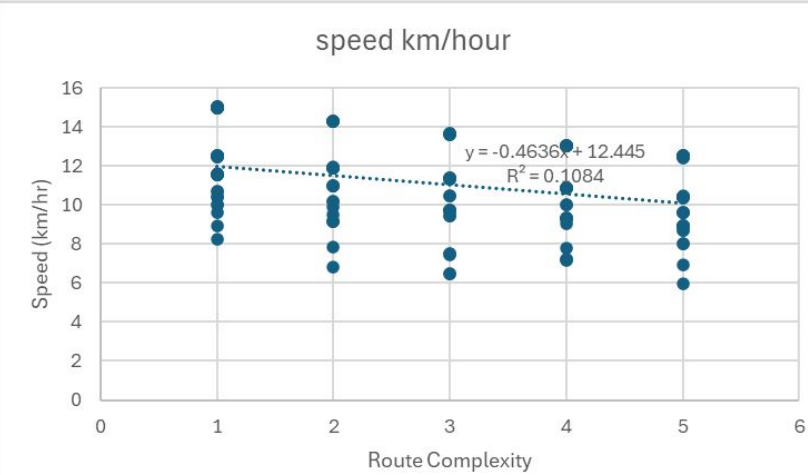


Data Exploration & Model Training

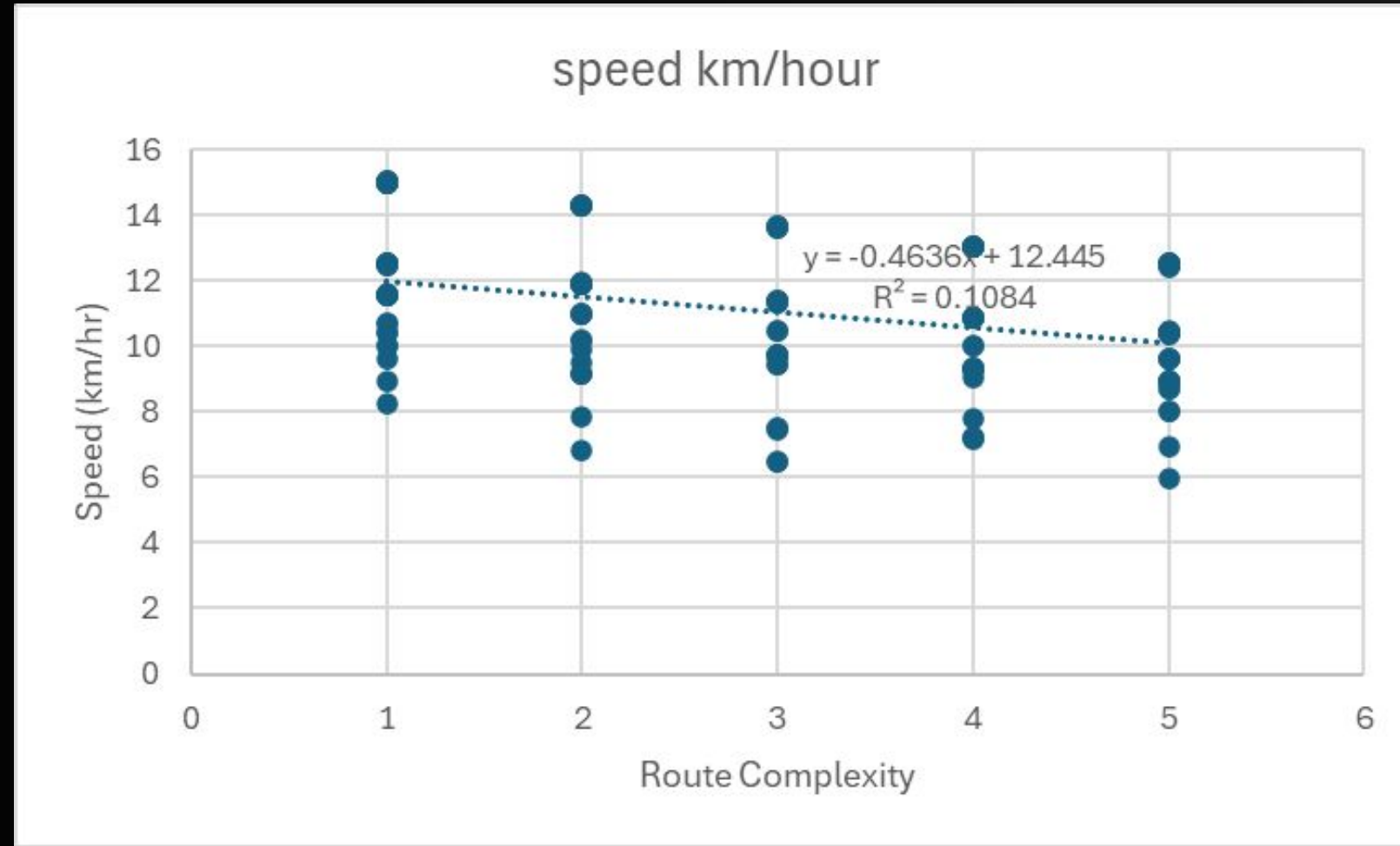
- *Created graphs for our data on Excel and through python on Colab*
- *Used the Excel add ons and training libraries on python to make models*

DATA ANALYSIS RESULTS

Package Weight & Route Complexity

	CORRELATION COEFFICIENT (R SQUARED(	MODEL EQUATION	DESCRIPTION	VISUAL
PACKAGE WEIGHT	$R^2 = 0.0041$	$y = 0.1029x + 10.732$	As package weight increases, speed increases	
ROUTE COMPLEXITY	$R^2 = 0.10842$	$y = -0.4636x + 12.445$	As route complexity increases, speed decreases very linear correlation	

Route Complexity

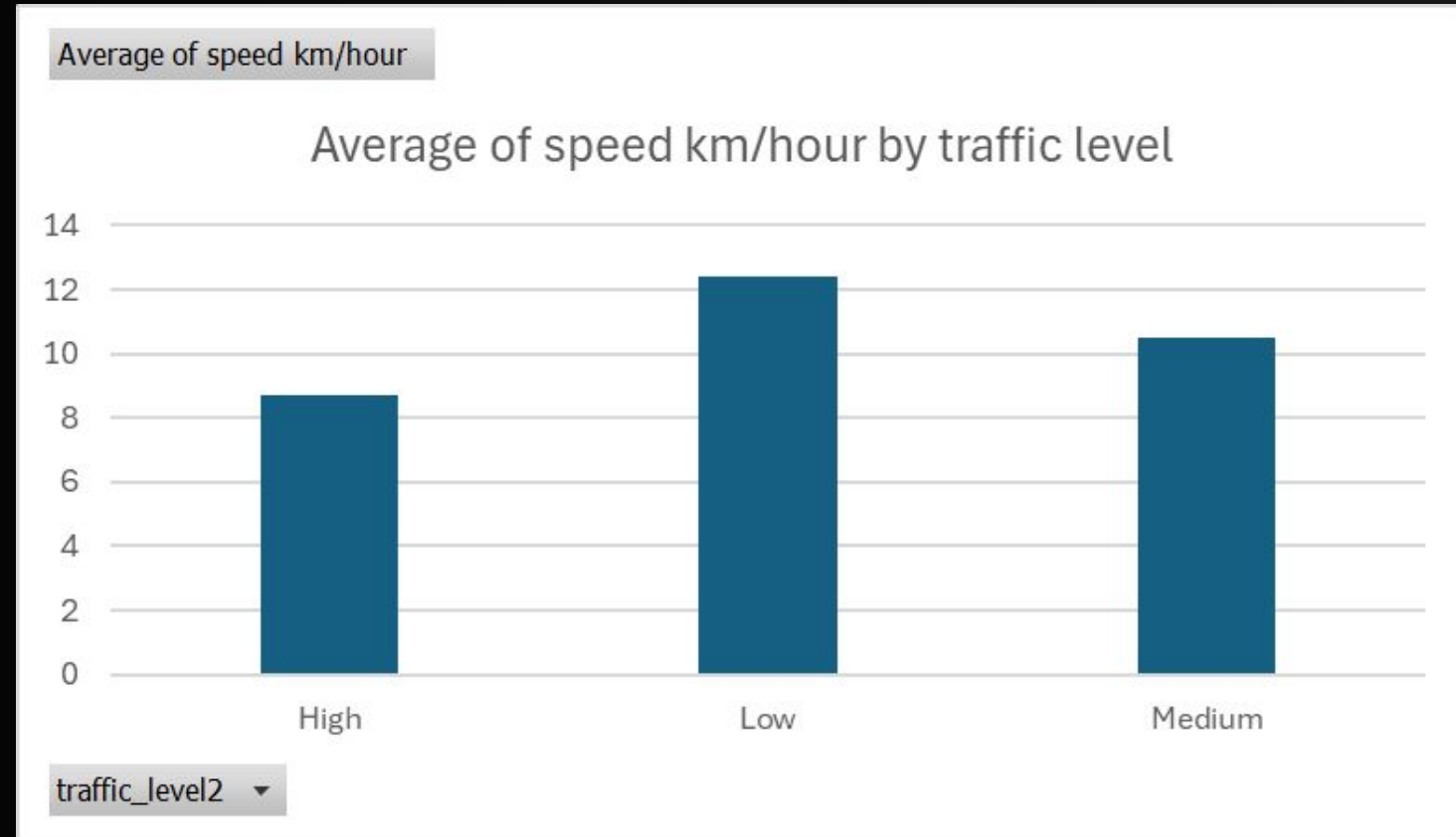


For each route complexity level we see a linear relationship with slope -0.4636

Route complexity is quite impactful on delivery speed if there are large leaps in complexity, if the level is only ± 1 , then weather and traffic is a larger issue to worry about.



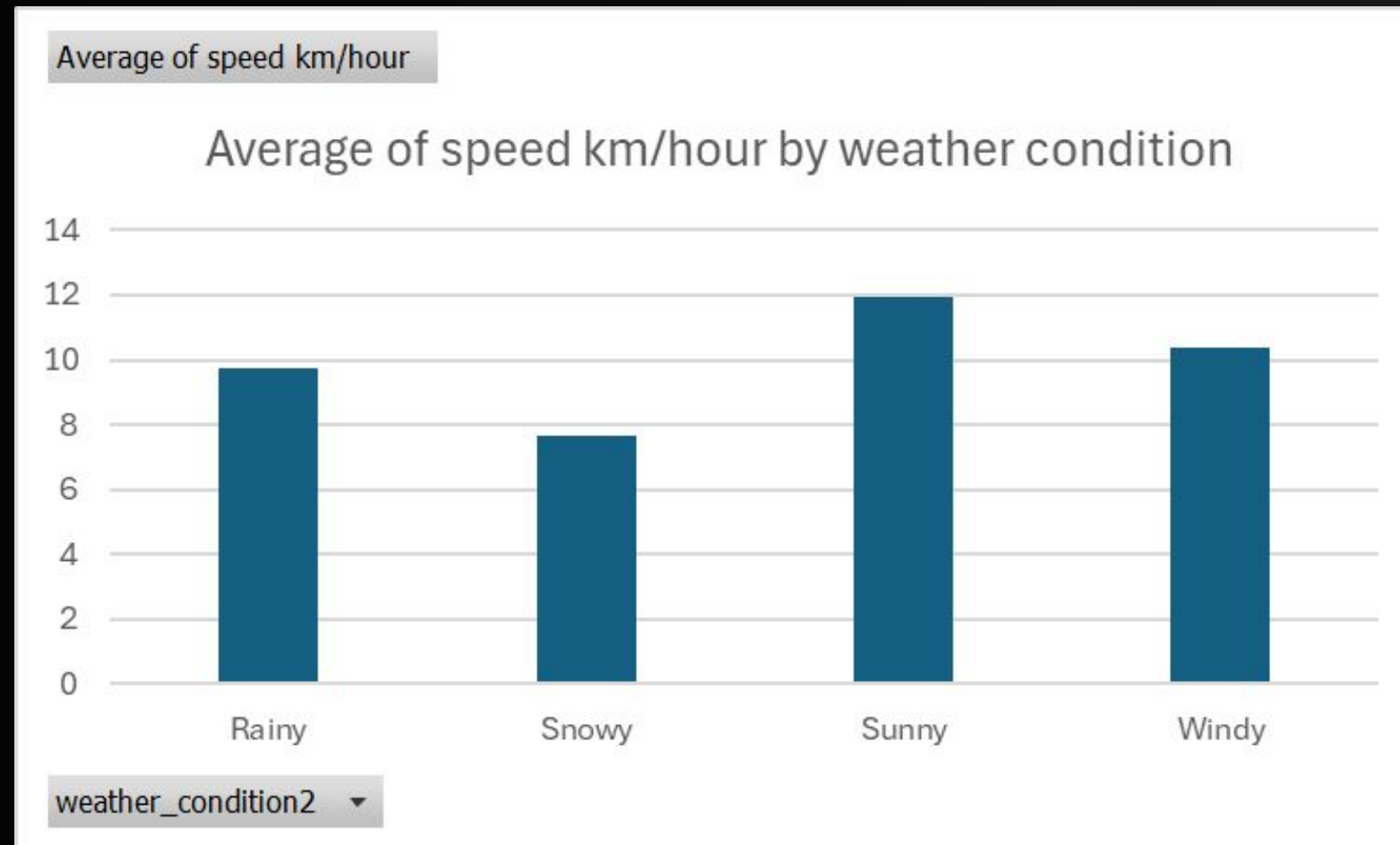
Traffic Level



- High traffic has an average speed of 8.71 km/h
- Low traffic has an average speed of 12.43 km/h



Weather

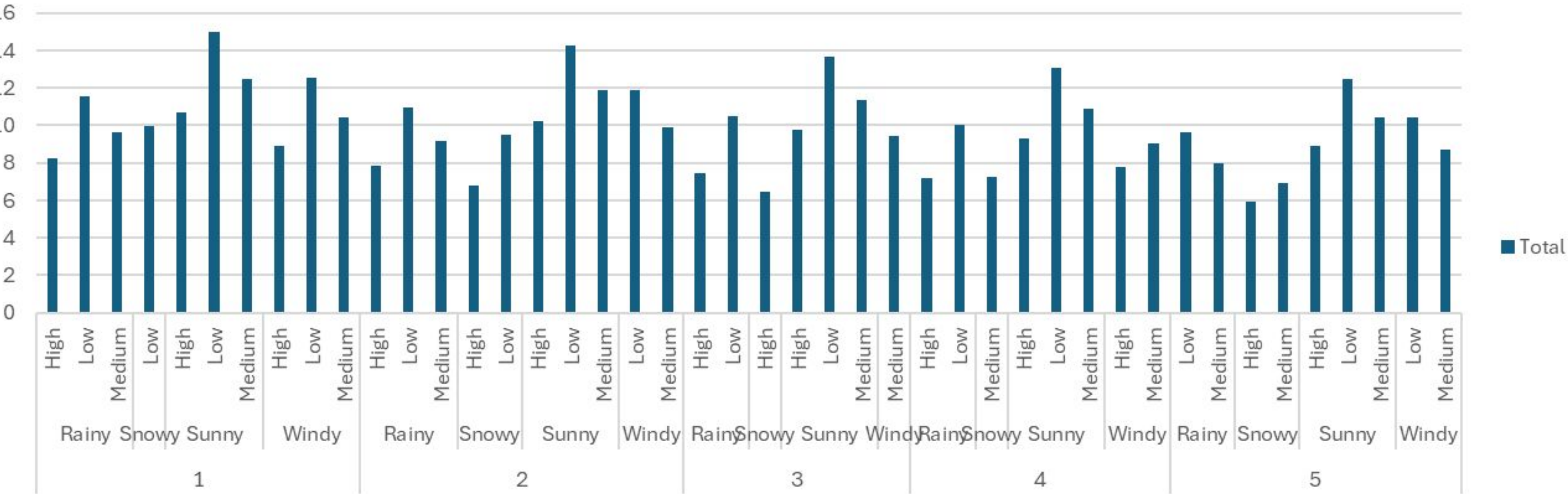


- Sunny weather has an average speed of 11.96 km/h (highest average speed)
- Snowy weather has an average speed of 7.67 km/h (lowest average speed)



Average of speed km/hour

Total



route_complexity ▼ weather_condition ▼ traffic_level ▼

+ -

Package Weight, Route Complexity, & Battery Level

Effect on Speed

R² values

Model Performance

R-squared: 0.0064

MSE: 0.0011

Model Values

Intercept: 0.2292
package_weight: -0.0013
route_complexity: -0.0079
battery_level: -0.0003

Value Comparison

	Actual	Predicted
73	0.217284	0.180190
18	0.173370	0.173497
118	0.249880	0.204884
78	0.198400	0.187237
76	0.227080	0.174682

COMPARISON

Package Weight	Route Complexity	Battery Level	Model Prediction	Actual Value
2.15	4	58.4	0.217284	0.180190
0.68	5	61.5	0.173370	0.173497
2.12	1	54.1	0.249880	0.204884
4.51	2	80.4	0.198400	0.187237
4.88	3	97.1	0.227080	0.174682
1.01	5	80.8	0.148846	0.168197
3.70	1	60.9	0.208223	0.201080

Distance, Route Complexity, & Delivery Time

Model Performance

R-squared: 0.7954
MSE: 0.0002

Intercepts:
0.1866

Coefficients:

distance_km: 0.0147
route_complexity: -0.0024
delivery_time_minutes: -0.0025

	Actual	Predicted
73	0.217284	0.209959
18	0.173370	0.175018
118	0.249880	0.232618
78	0.198400	0.206954
76	0.227080	0.225946

The Model improves when I only use 10% of the data for testing instead of 20%.

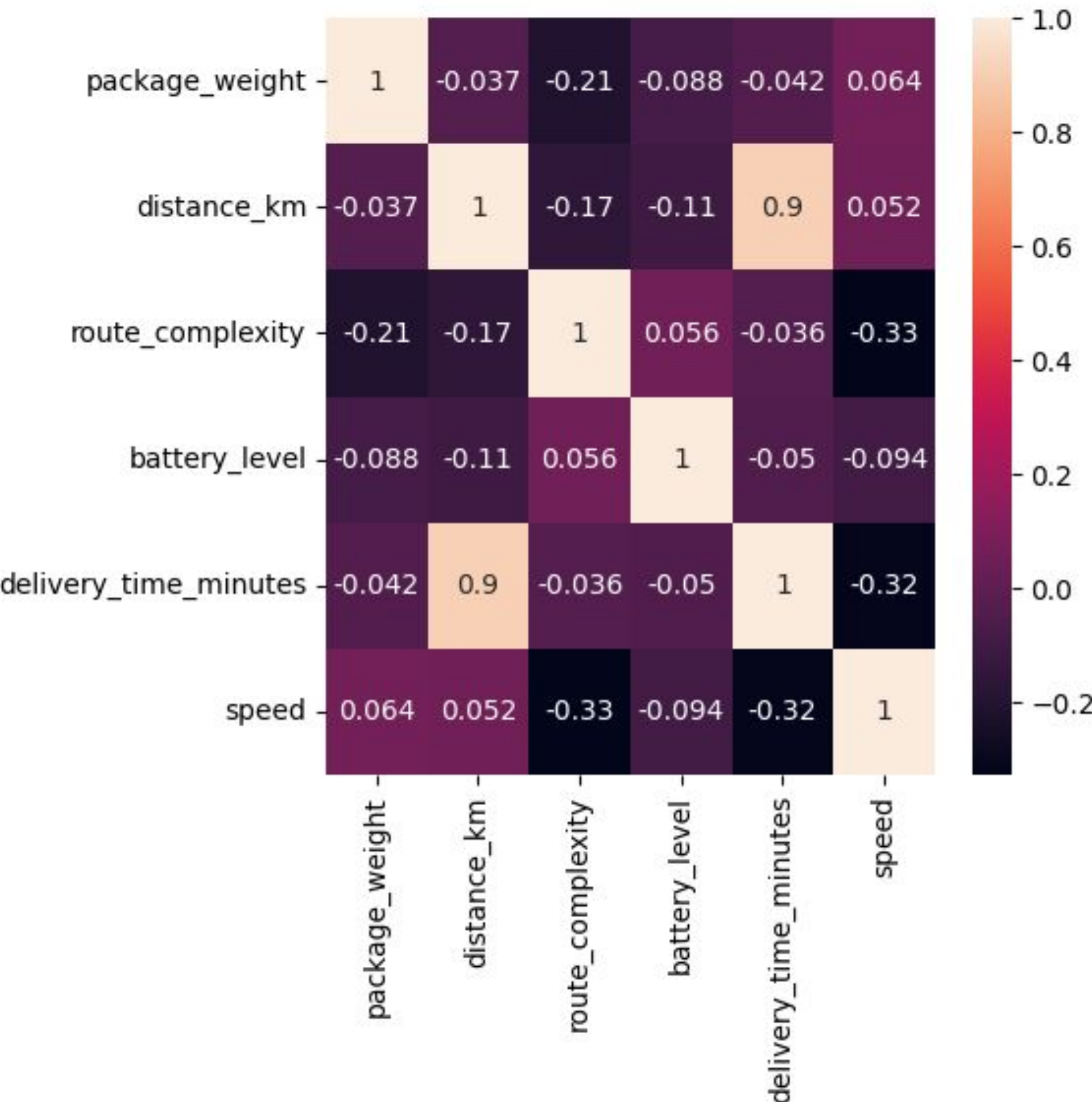
New Values:

R-squared: 0.8384
Intercept: 0.1896

Coefficients:

distance_km: 0.0147
route_complexity: -0.0027
delivery_time_minutes: -0.0026

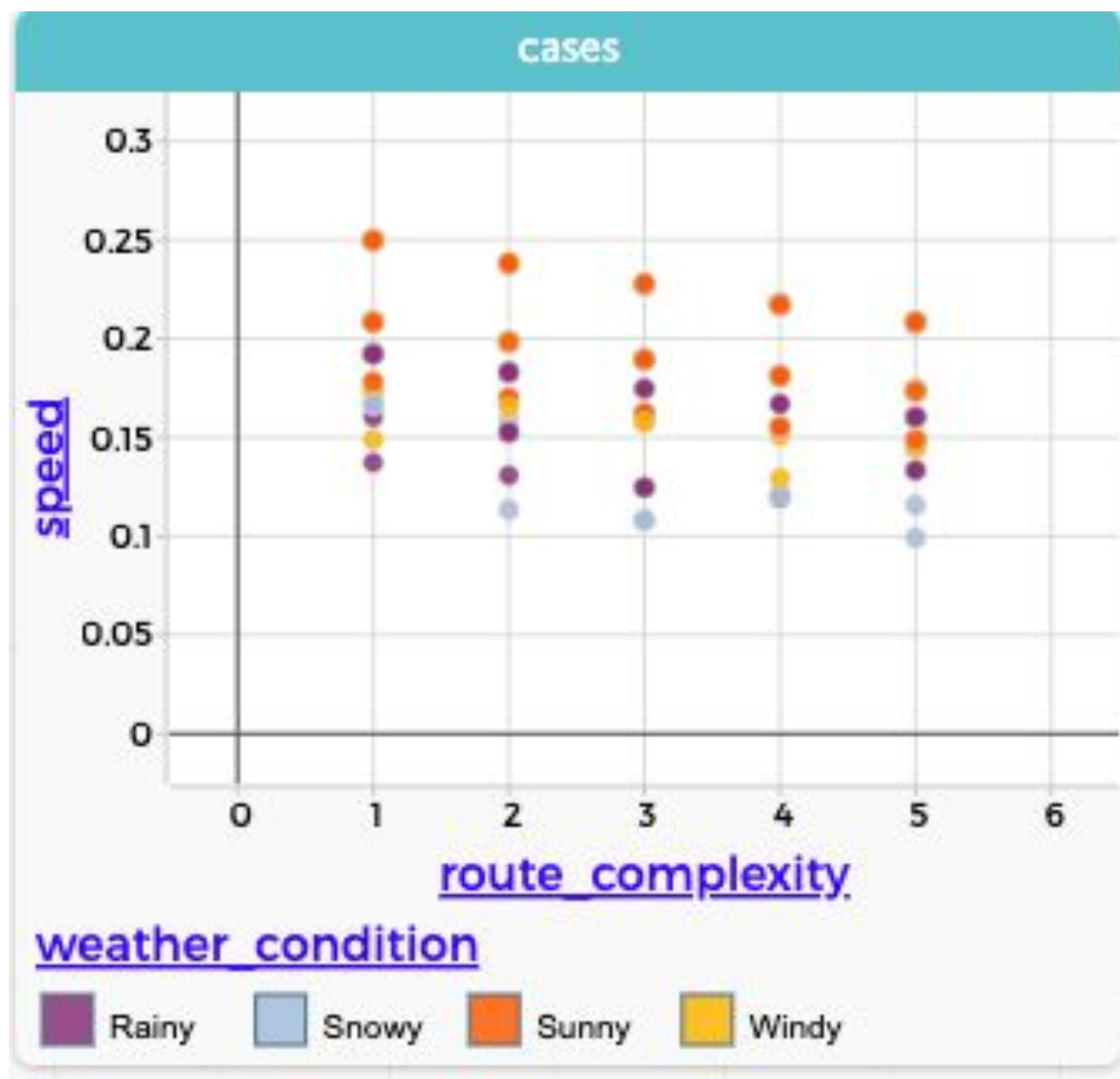
SEABORN HEATMAP



Correlation Matrix:

	package_weight	distance_km	route_complexity
package_weight	1.000000	-0.036935	-0.212507
distance_km	-0.036935	1.000000	-0.167648
route_complexity	-0.212507	-0.167648	1.000000
battery_level	-0.088456	-0.107142	0.056031
delivery_time_minutes	-0.042151	0.903250	-0.036051
speed	0.063708	0.051978	-0.329281

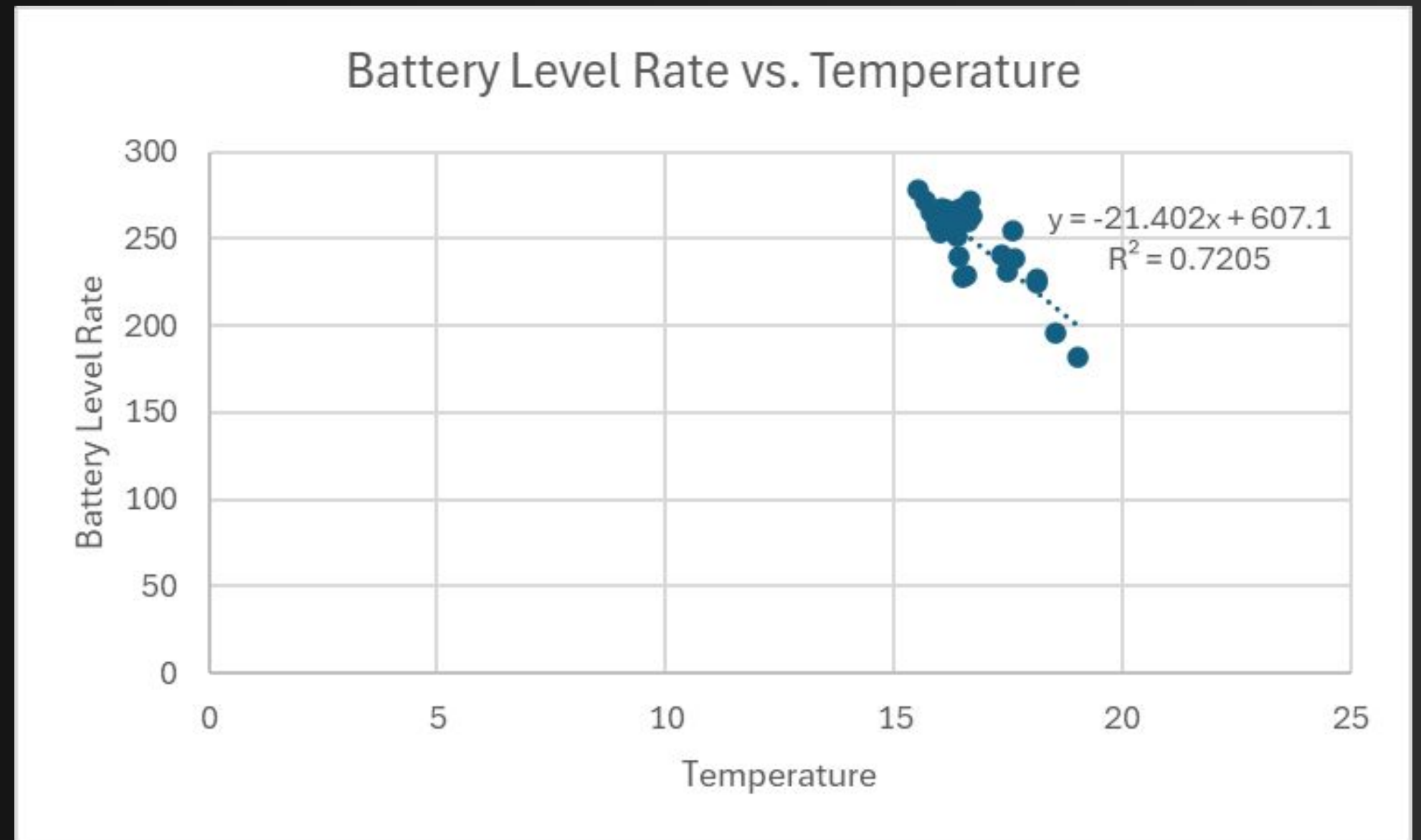
	battery_level	delivery_time_minutes	speed
package_weight	-0.088456	-0.042151	0.063708
distance_km	-0.107142	0.903250	0.051978
route_complexity	0.056031	-0.036051	-0.329281
battery_level	1.000000	-0.049733	-0.094406
delivery_time_minutes	-0.049733	1.000000	-0.318277
speed	-0.094406	-0.318277	1.000000



Battery Level and Temperature

As temperature increases, battery level decreases more quickly

$$y = 607.1033 - 21.4025x$$
$$R \text{ Squared} = 0.720524$$



Findings

- Package weights have little/no correlation with speed of the drone, so companies do not need to worry about how heavy a package is
- Route complexity doesn't slow down drones much
- Snowy weather slows the drone down the most
- Sunny weather allows the drone to travel the fastest
- However, high temperatures make the battery run out most quickly

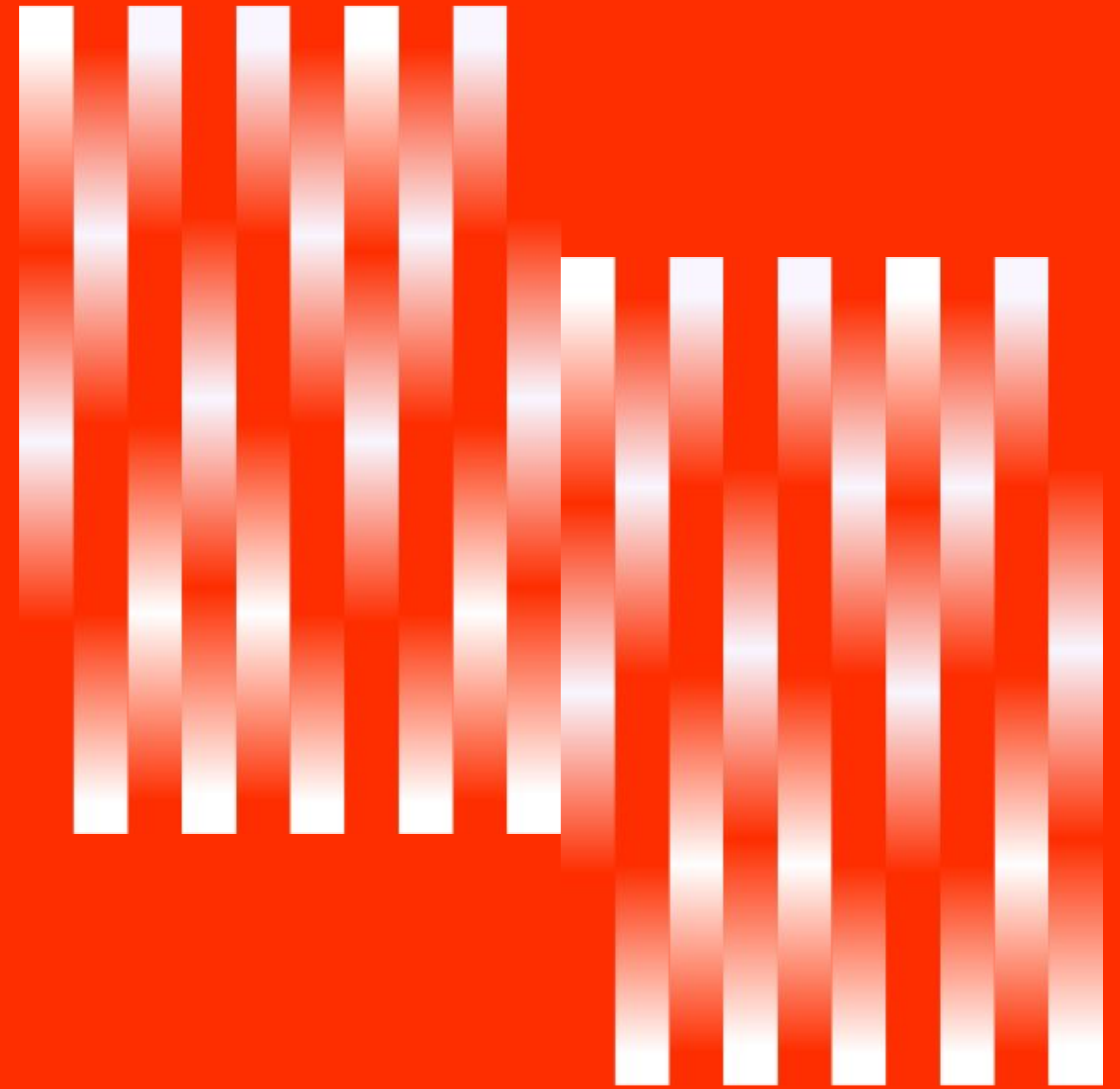




Limitations & Further Research

- This data set was relatively small (only 150 rows of data)
- Research should be done in other months of the year and different days (not just January 31st) to evaluate different weather conditions
 - More research should be done to find how each condition (package weight, route complexity, weather, temperature, and traffic affect speed of a drone)
- Research on unsuccessful drone deliveries would show the most important conditions for accomplishing a drone delivery

THANK YOU



REFLECTION

We are most proud of how we were able to communicate and come together on this project. At the start, we were each picking a tool we wanted to use (such as Excel, Colab, and Codeap), but by the final week we were able to bring everything together to present our analysis.

CREDITS

This presentation template is free for everyone to use, thanks to the following:



for this presentation template

Pexels, Pixabay, Sketchify

for the photos, graphics, and elements

Happy designing!

<i>Regression Statistics Speed Prediction</i>	
Multiple R	0.329364981
R Square	0.108481291
Adjusted R Square	0.090162413
Standard Error	1.255475133
Observations	150

package weight, distance,
route complexity

<i>Regression Statistics Delivery Time</i>	
Multiple R	0.910970801
R Square	0.8298678
Adjusted R Square	0.826371933
Standard Error	10.91599719
Observations	150

RESOURCE PAGE

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Fonts

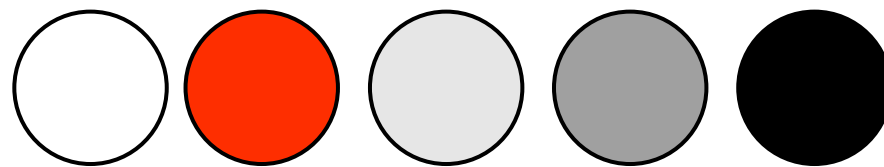
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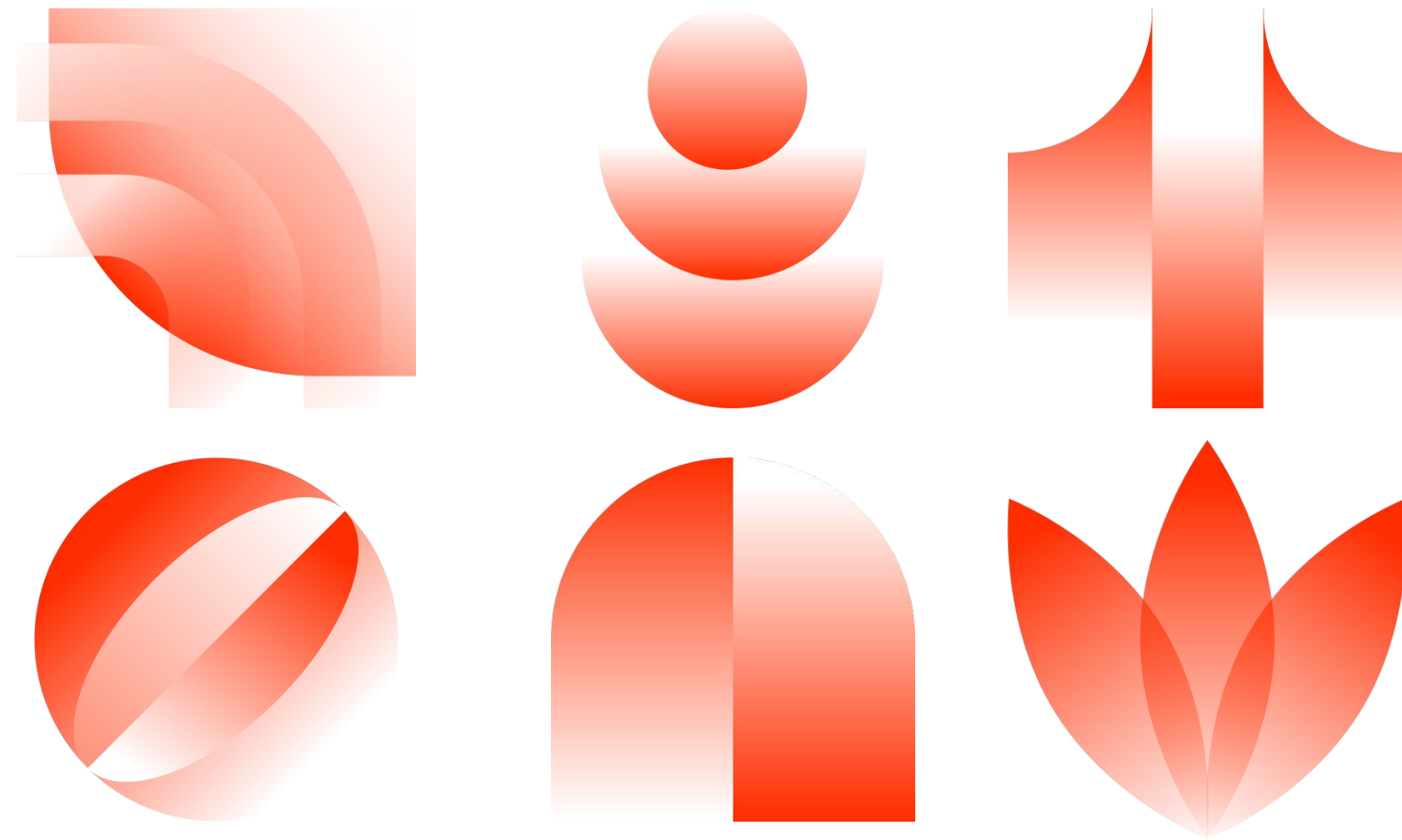
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Colors

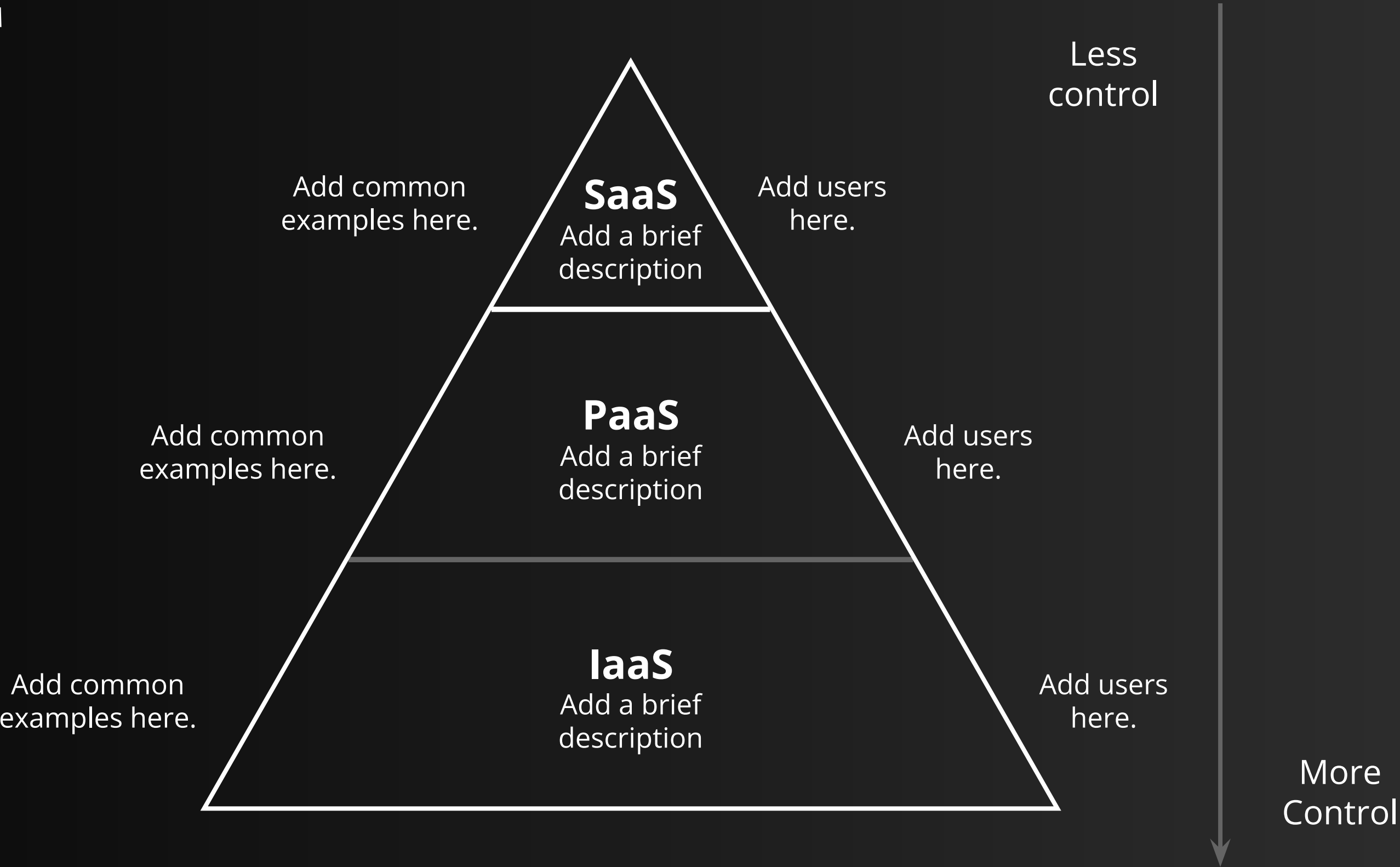


#FFFFFF #FE2E00 #E6E6E6 #A0A0A0 #000000

Design Elements / Icons



TYPES OF CLOUD SERVICES



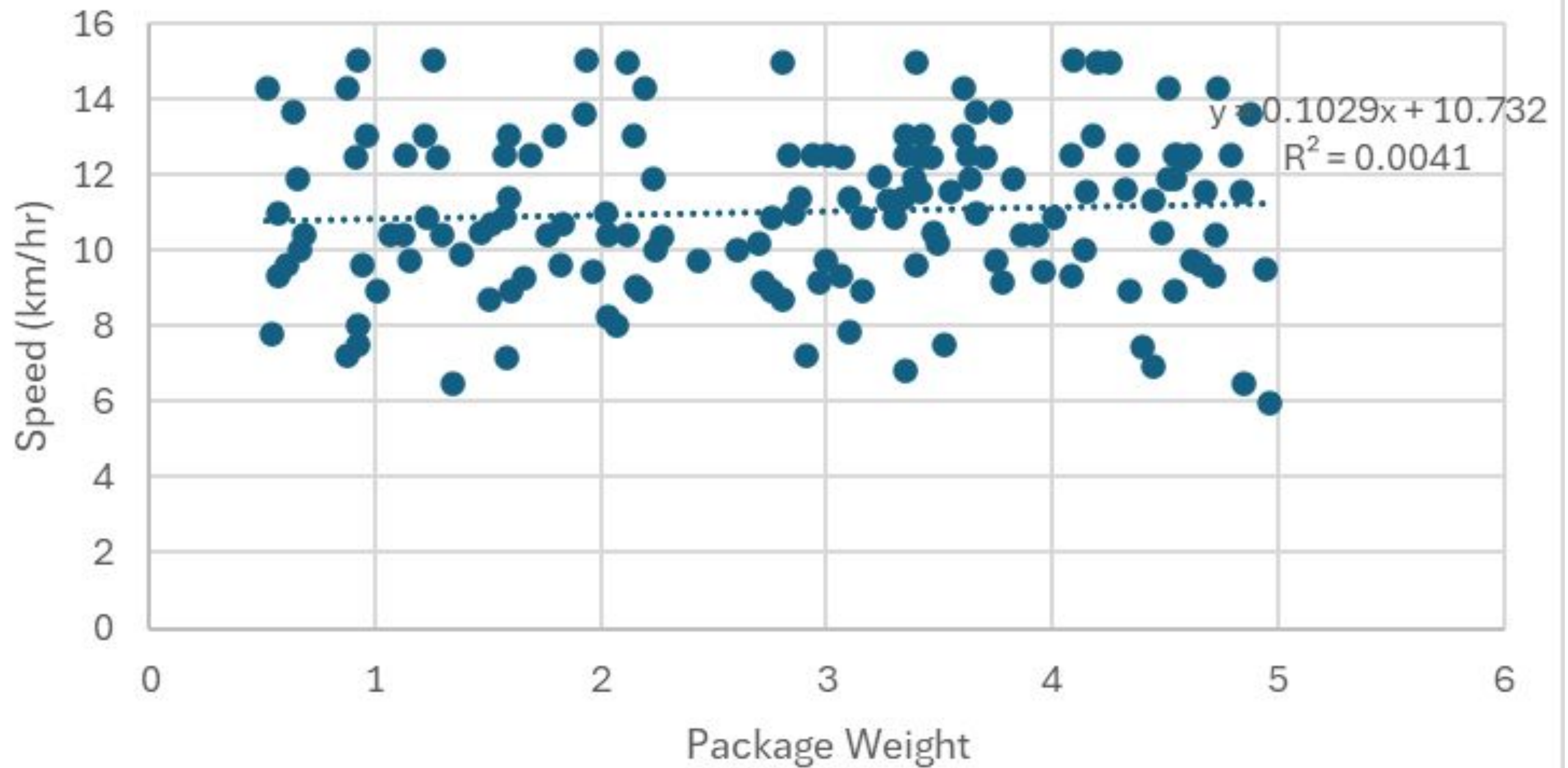
Regression Statistics

Multiple R	0.848837
R Square	0.720524
Adjusted R Square	0.710173
Standard Error	12.32769
Observations	29

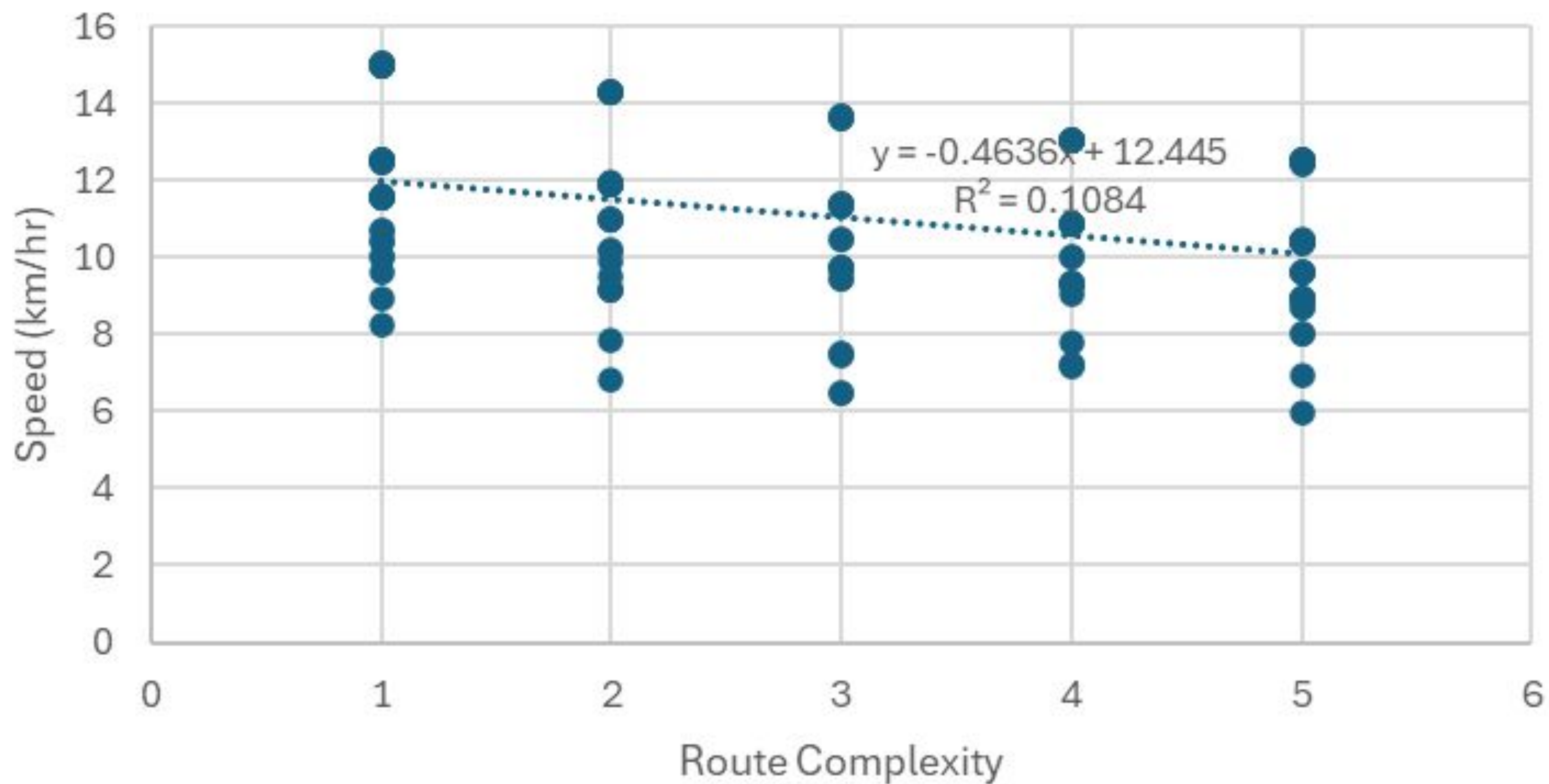
Between temperature and battery level

$$y = 607.1033 - 21.4025x$$

Package Weight vs. Speed



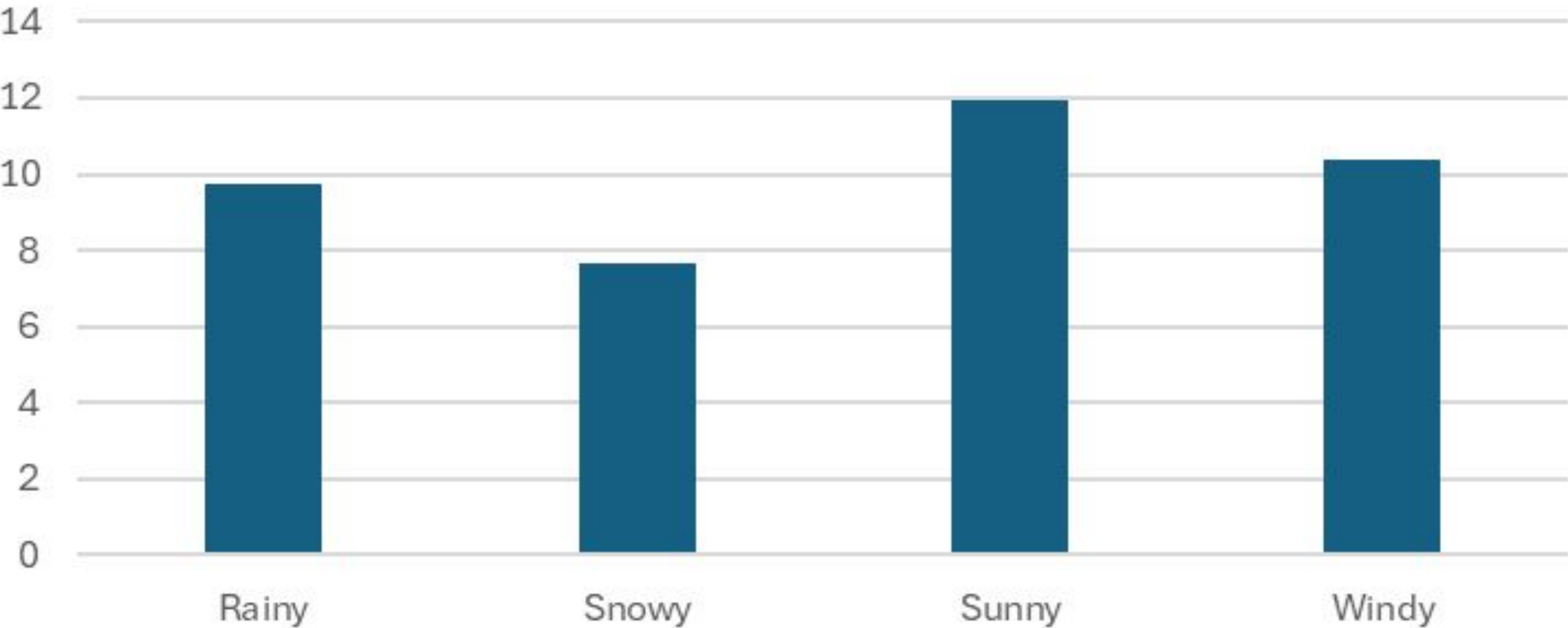
speed km/hour



Average of speed km/hour

weather_condition2	Average of speed km/hour
Rainy	9.71757753
Snowy	7.669479554
Sunny	11.96058825
Windy	10.35864283

Average of speed km/hour by weather condition

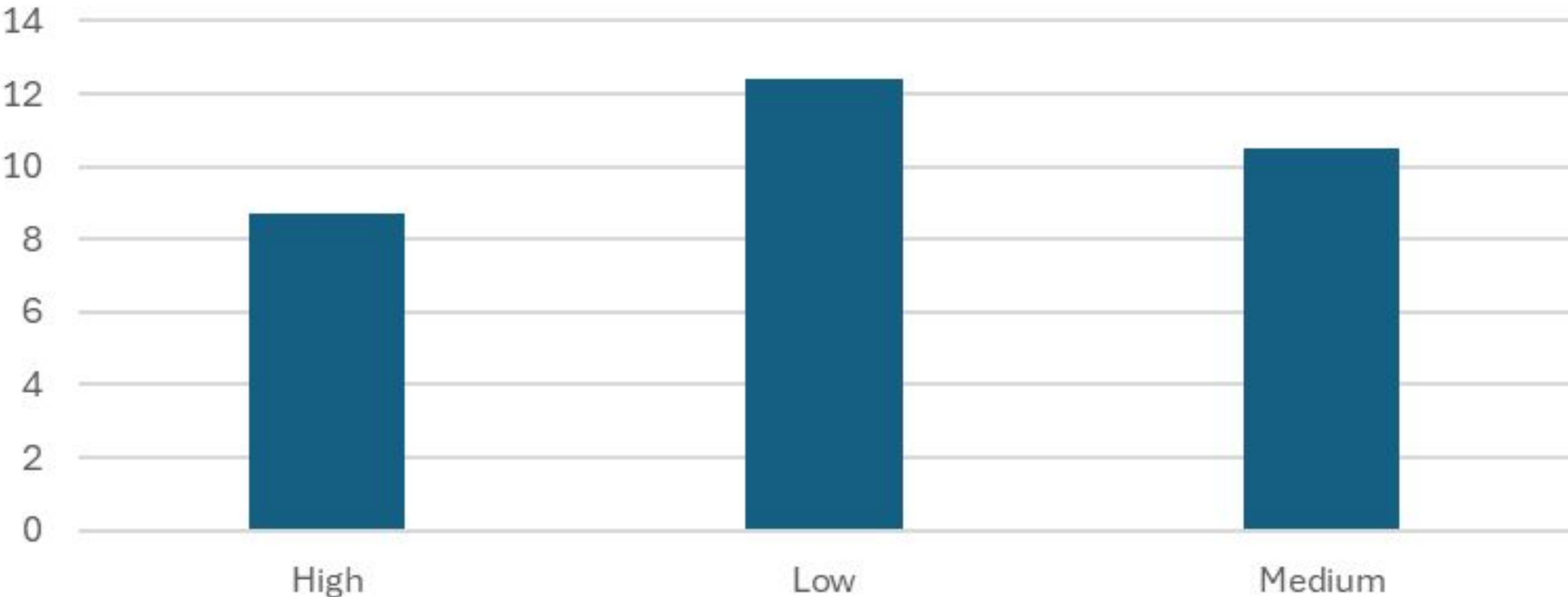


weather_condition2 ▼

Average of speed km/hour

traffic_level2	Average of speed km/hour
High	8.711826868
Low	12.42960915
Medium	10.47168419

Average of speed km/hour by traffic level



traffic_level2 ▼

Battery Level vs. Speed

